



Use Earth Observation, AI, geospatial data and your ideas to develop innovative solutions for the 2026 Nepal flash-flood and debris-flow event.

	01 WHEN THE LANDSCAPE CHANGES Mapping Post-Flood LULC Alterations Detect and quantify land-use and land-cover changes caused by the 2026 flood and debris-flow event.	
	02 WHEN HERITAGE IS UNDER THREAT Assessing Flood Damage to Cultural Sites Identify and assess potential impacts on cultural heritage sites, temples and historic structures.	
	03 GIVE THE RIVER ROOM Detecting River Channel Constriction and Flood Implications Identify where human development has constricted river channels and assess its influence on flood behaviour.	
	04 FROM GLACIER TO AVALANCHE Tracing the Root Causes of Cascading Hazards Investigate environmental and physical drivers of glacier melt, glacier instability and ice-rock avalanches.	
	05 AFTER THE FLOOD Mapping WASH Disruptions and Community Needs Assess disruptions to water, sanitation and hygiene services and identify communities requiring priority intervention.	
	06 ONE RIVER, TWO COUNTRIES Understanding Transboundary Flood Impacts Analyse how floodwaters and impacts propagated from Nepal to downstream areas in Bihar, India.	
	07 BEYOND THE KNOWN An Open Challenge for Disaster Innovation Explore any other significant, innovative and meaningful problem related to the 2026 event.	



Rasuwa Flood Portal

An open, living 3D map of the 2026 Bhotekoshi flood

Theme 07 · Beyond the Known

An open challenge for disaster innovation

GeoDev

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Project Overview



Summary

On 26 August 2026, a rock and ice collapse near Langtang Lirung sent a debris flood about 72 km down the Bhotekoshi–Trishuli valley in Rasuwa, Nepal. Information about the damage was scattered across agencies, volunteers and satellite platforms. The Rasuwa Flood Portal brings it into one open 3D web map: flood extent, impact layers, landslides and Sentinel-2 before/after imagery. It shows ~145 km of damaged road and 190 exposed settlements along a 45 km² flood extent.



Innovation statement

A living portal, not a one-off map. Anyone can add a dataset by editing one config file, with no code. It puts impact data, a 3D Sentinel-2 before/after swipe (True Color, False Color, NDVI) and a flood-surge elevation profile together in one free, open tool.



BASIC Information

Study Area

- Bhotekoshi–Trishuli river corridor, Bagmati & Gandaki provinces, Nepal
- From the Langtang Lirung glacier source to the downstream Trishuli
- 24 affected municipalities in 8 districts, incl. Rasuwa, Nuwakot, Dhading

Data

- Sentinel-2 L2A, 12 Aug & 21 Sep 2026 (Copernicus)
- Vantor post-event imagery (HOT)
- OSM-based impact layers: roads, settlements, facilities
- BGS landslide inventory
- HOT/HDX, Overture, NAXA; AWS terrain

Method

- Collect & check open datasets
- Clean and reproject to GeoJSON
- Register each layer in one config file
- Render in 3D with MapLibre
- Sentinel-2 swipe + NDVI for change

Output

- Live, open web portal
- 40 toggleable map layers
- Before/after swipe, 3D tour, elevation profile
- Layer stats + downloadable GeoJSON

Impact

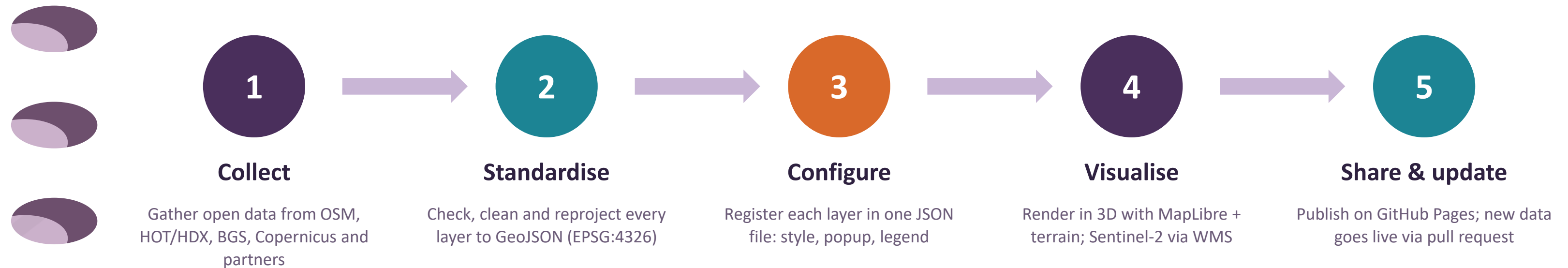
- Responders: see which roads, bridges and settlements to reach first
- Mappers & researchers: one shared base map
- Media & public: understand the event visually

SDG Connection

- **SDG 11.5** reduce disaster losses
- **SDG 13.1** resilience to climate hazards
- **SDG 9.1** resilient infrastructure
- **SDG 17** open-data partnerships



Method

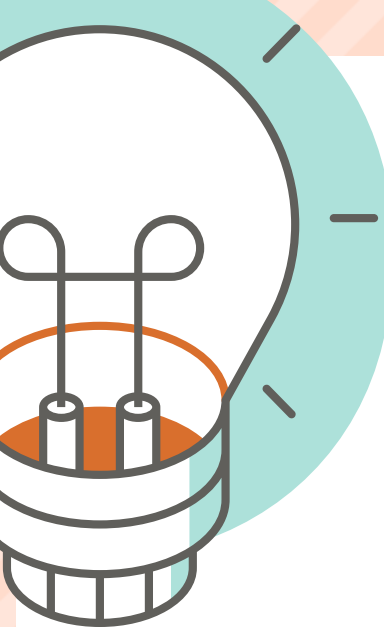


Adding a layer = a few lines of JSON

```
{
  "id": "road_damaged",
  "label": "Damaged Road Network",
  "type": "line",
  "data": "data/.../Road_damaged.geojson",
  "paint": { "lineColor": "#d62728" },
  "active": true
}
```

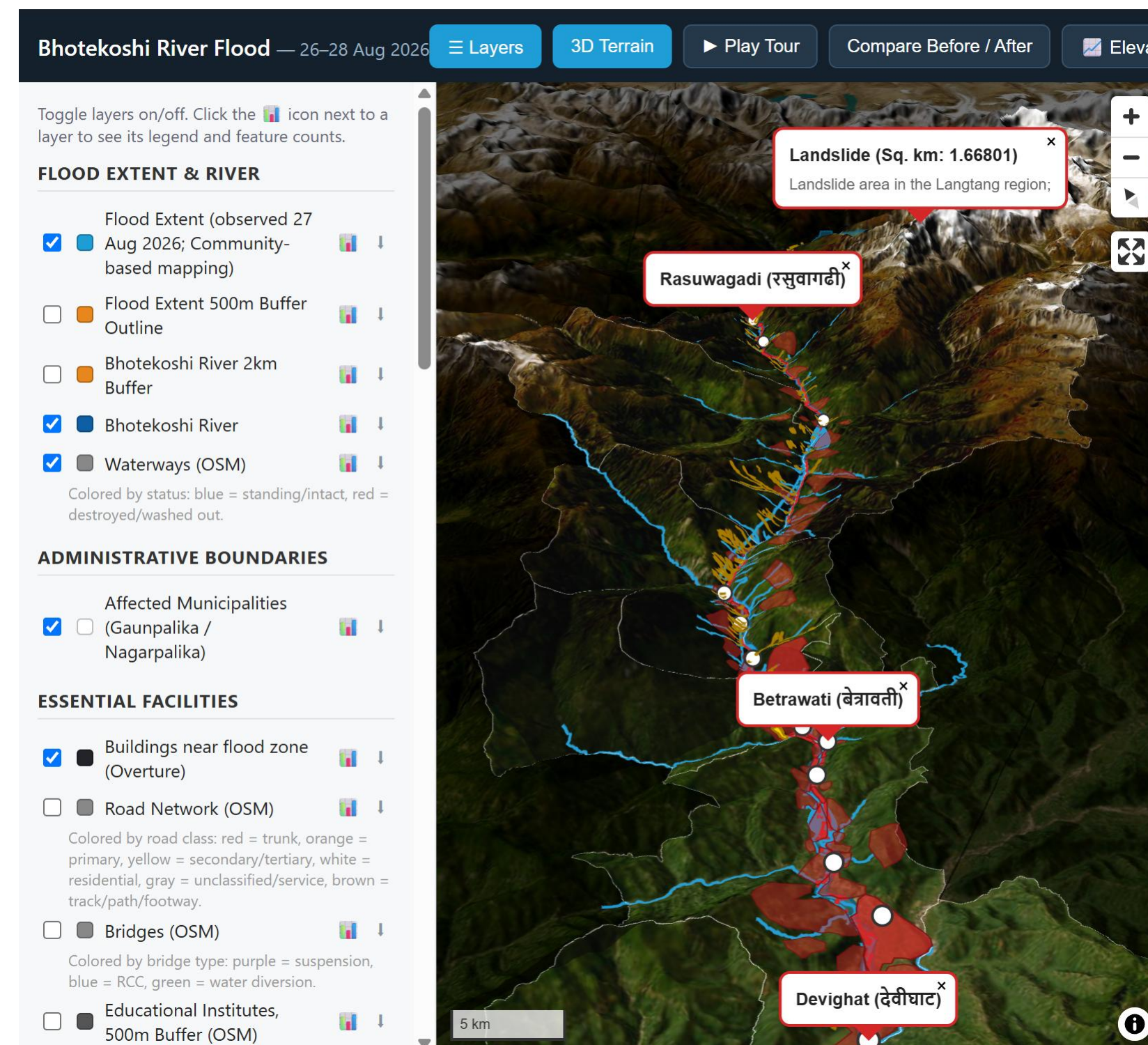
Built with open tools

- **MapLibre GL JS** 3D map, terrain, swipe compare
- **Copernicus Sentinel Hub** Sentinel-2 True/False Color, NDVI
- **GeoJSON + layers.json** all data and styling in one config
- **Python** data prep and the elevation profile
- **GitHub Pages** free hosting, open contributions



Results

Impact at a glance



~145 km

damaged road

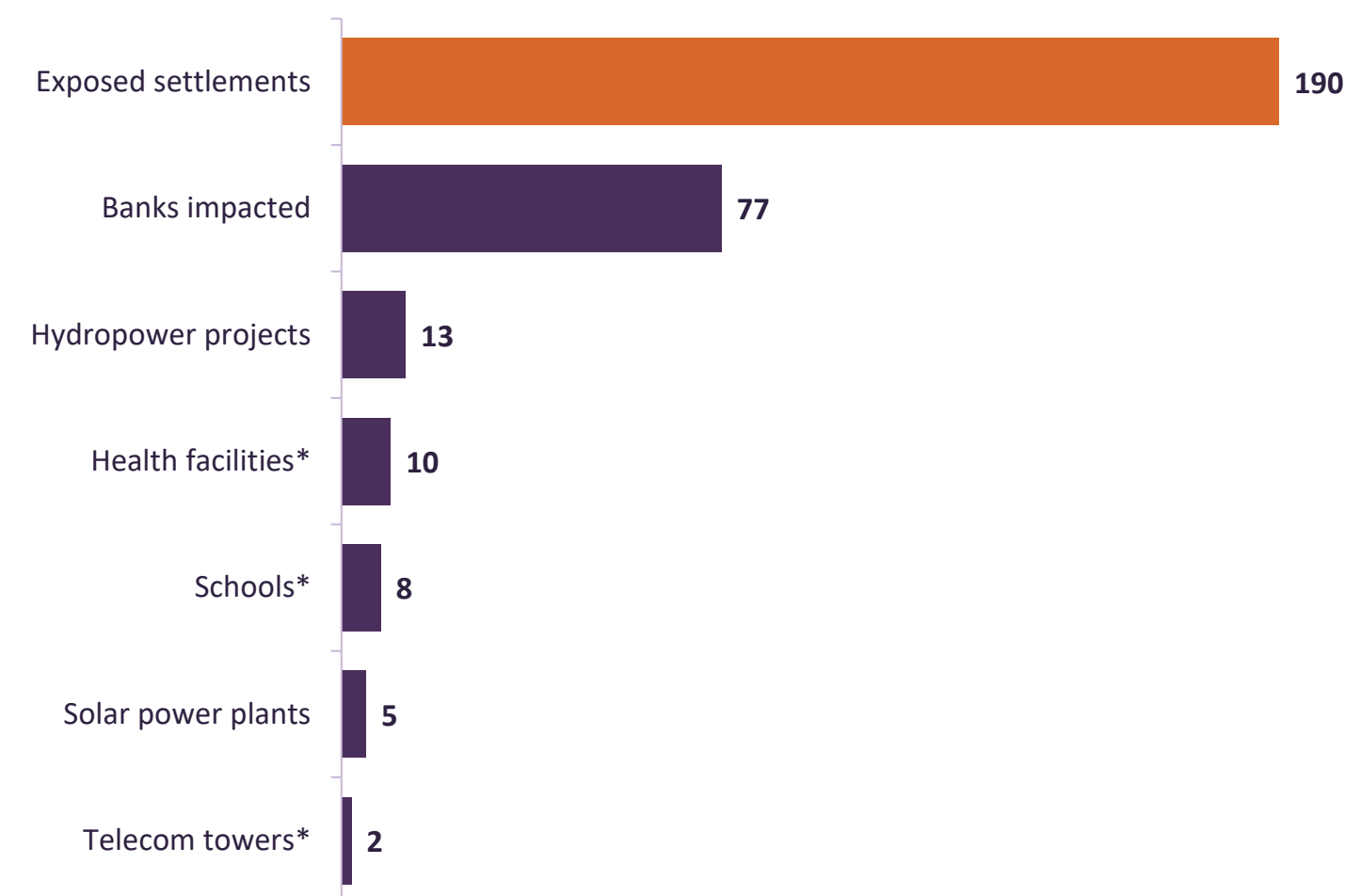
~163 km

road at risk

45 km²

flood extent

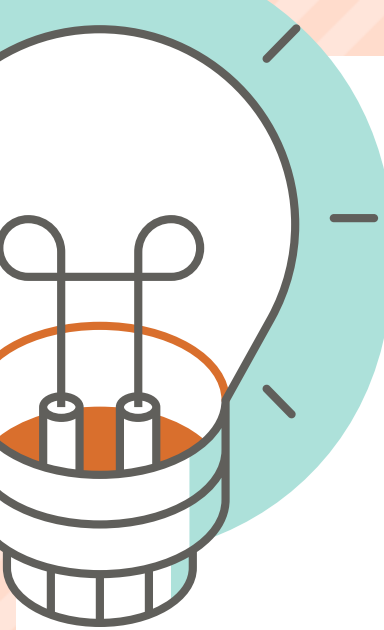
Assets in the flood corridor (count)



* within the mapped flood extent. Source: OSM-based impact layers on the portal; road lengths summed from segments.

Takeaway: with ~145 km of road damaged and 163 km more at risk, the map shows responders which settlements and facilities to reach first.





Results

Before vs after, Sentinel-2



Takeaway: one swipe over 3D terrain shows how the river channel and vegetation changed between 12 Aug and 21 Sep, with no GIS skills needed.





References



In Same study area

- **NDRRMA / OSM (2026).** Flood impact layers for the Bhotekoshi–Trishuli corridor: roads, settlements, facilities.
- **HOT OSM (2026).** Nepal Floods 2026 damage assessment. HDX dataset “hot_flood_npl” (data.humdata.org)

Similar study

- **Shugar, D.H. et al. (2021).** A massive rock and ice avalanche caused the 2021 disaster at Chamoli, Indian Himalaya. *Science* 373, 300–306.
- **ICIMOD (2021).** The Melamchi flood disaster: cascading hazard and the need for multihazard risk management.

Similar solution

- **HOT OSM uMap (2026).** Nepal Floods 2026 focus area & tasking map: umap.hotosm.org
- **Copernicus Browser / EMS Rapid Mapping.** Satellite before/after viewing and emergency maps.

Use / Scope of Knowledge-Driven AI

Where AI is used today

- AI-assisted building-damage mapping in the HOT datasets, validated by human mappers before it reaches the portal
- AI coding assistant used to speed up portal development

Where knowledge-driven AI adds value next

- Sentinel-2 change detection: NDVI/NDWI differencing + an ML classifier to flag new debris and scoured banks
- Priority ranking from expert rules, e.g. hazard × exposure (a school within 500 m of a damaged road)
- Plain-language Q&A over the layers: “Which health posts lost road access?”

Why it fits: data is large and time-critical. Limits: monsoon cloud and scarce ground truth, so people stay in the loop.



CONCLUSIONS

The Rasuwa Flood Portal turns scattered, multi-source flood data into one open 3D map that anyone can explore. The impact layers show ~145 km of damaged road and 190 exposed settlements along a 45 km² flood extent, and the Sentinel-2 swipe makes landscape change visible without GIS skills. Because layers are added through one config file, the portal keeps improving as new data arrives.

Initial objectives

- ✓ **Bring all event data into one place** — achieved: 40 layers from 10+ sources
- ✓ **Make the impact visible** — achieved: roads, settlements, facilities, landslides
- ✓ **Show before/after change clearly** — achieved: 3D Sentinel-2 swipe with NDVI
- 🔄 **Keep it open and community-driven** — ongoing: contributions via GitHub



Graphical Abstract



Problem

Flood data scattered across agencies, volunteers and satellites



Data

Sentinel-2
Vantor imagery
OSM impact layers
BGS landslides
Terrain DEM



People

Responders
Mappers & researchers
Media & public



45 km² flood extent 145 km damaged road 190 exposed settlements 915 landslides 40 open layers

Thanks!

Live portal

iamtekson.github.io/rasuwa-flood

Code & data

github.com/iamtekson/rasuwa-flood

Contact

[email / LinkedIn]



Scan to open the portal

